Stock Portfolio Analyzer - User Guide

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Introduction

Stock Portfolio Analyzer is a Streamlit web application that helps investors analyze stock performance, calculate risk-return metrics, and optimize portfolio allocation using Modern Portfolio Theory. The application provides interactive visualizations and data analysis tools to make informed investment decisions.

Project Overview

Purpose and Capabilities

The Stock Portfolio Analyzer application allows users to:

- Track historical stock price performance for multiple securities
- Compare stocks using percentage change analysis
- Analyze technical indicators like moving averages
- Examine trading volume patterns
- Visualize the distribution of daily returns
- Assess risk-return profiles for individual stocks
- Generate correlation matrices to understand relationships between stocks
- Calculate the efficient frontier for portfolio optimization
- Determine optimal portfolio allocation for maximum Sharpe ratio

Technology Stack

- **Streamlit**: Web application framework for data applications
- **Pandas**: Data manipulation and analysis
- **yfinance**: Yahoo Finance API for stock data retrieval
- **Matplotlib/Seaborn**: Data visualization
- **NumPy**: Numerical computations for portfolio optimization

Setup and Installation

Prerequisites

To run the Stock Portfolio Analyzer, you need:

- Python 3.7+
- pip (Python package installer)

Required Packages

```bash
pip install streamlit pandas yfinance matplotlib seaborn numpy
```

Running the Application

1. Save the code to a file named `stock_portfolio_analyzer.py`
2. Open a terminal/command prompt
3. Navigate to the directory containing the file
4. Run the application with:

```bash
streamlit run stock_portfolio_analyzer.py
```

5. The application will open in your default web browser

Application Flow

The Stock Portfolio Analyzer follows this logical flow:

1. **Configuration**: The application starts by setting up the page configuration and UI elements

2. **User Input Collection**: The sidebar collects user inputs like date range, ticker symbols, and analysis options
3. **Data Retrieval**: Upon clicking "Run Analysis", stock data is downloaded via yfinance API
4. **Data Processing**: The application:
   - Calculates daily returns and percentage changes
   - Computes moving averages and other technical indicators
   - Derives risk-return metrics
   - Simulates random portfolios for optimization
5. **Visualization Generation**: Creates charts and tables based on user-selected analysis options
6. **Result Display**: Shows the analysis results in the main content area

User Interface Guide

Sidebar Controls

Portfolio Settings

- **Start Date & End Date**: Define the historical period for analysis
- **Stock Selection**: Enter stock symbols (one per line)
- **Risk-Free Rate**: Set the risk-free rate percentage for Sharpe ratio calculations
- **Number of Portfolio Simulations**: Control the accuracy of the efficient frontier

Technical Analysis

- **Short-term MA Window**: Set the period for the short-term moving average (default: 50 days)
- **Long-term MA Window**: Set the period for the long-term moving average (default: 200 days)

Analysis Options

Toggle checkboxes to show/hide different analyses:

- Price Charts
- Percentage Change
- Moving Averages
- Volume Analysis
- Returns Distribution
- Risk-Return Analysis
- Correlation Matrix

- Efficient Frontier
- Optimal Allocation

Main Content Area

This is where analysis results appear after clicking "Run Analysis"

Analysis Features

Price Charts

Displays the closing price over time for all selected stocks, allowing for direct comparison of price movements.

Percentage Change

Shows relative performance of stocks from the starting date, normalized to a common baseline for fair comparison.

Moving Averages Analysis

For each stock, displays:

- The closing price
- Short-term moving average
- Long-term moving average

These technical indicators help identify trends and potential buy/sell signals when lines cross.

Volume Analysis

Shows trading volume over time, which can indicate market interest and validate price movements.

Returns Distribution

Visualizes the distribution of daily returns as a histogram with density curve, helping assess:

- Return volatility
- Skewness (asymmetry)
- Kurtosis (tail behavior)

Risk-Return Analysis

Provides a statistical summary and scatter plot showing:

- Expected annualized return
- Volatility (risk)
- Sharpe ratio

This allows comparison of stocks on a risk-adjusted basis.

Correlation Matrix

A heatmap showing correlation coefficients between stock returns, essential for:

- Understanding diversification benefits
- Identifying highly correlated securities
- Building balanced portfolios

Efficient Frontier

Plots the efficient frontier curve showing:

- Risk-return profiles of thousands of portfolio combinations
- Maximum Sharpe ratio portfolio (optimal allocation)
- Individual stock positions

Optimal Allocation

Shows the optimal portfolio weights that maximize the Sharpe ratio:

- Table with percentage weights
- Pie chart visualization
- Download option for saving results as CSV

Technical Details

Key Functions

`download_stock_data(tickers, start_date, end_date)`

- Downloads historical price data using yfinance
- Handles both single and multiple ticker scenarios
- Performs data cleaning and validation
- Returns a DataFrame with price data and daily returns
- Uses Streamlit's caching to improve performance

`calculate_portfolio_stats(stock_data, risk_free_rate, num_portfolios)`

- Calculates expected returns and volatility for individual stocks
- Computes the covariance matrix
- Simulates random portfolio weights
- Evaluates each portfolio's performance (return, volatility, Sharpe ratio)
- Identifies the optimal portfolio with maximum Sharpe ratio
- Returns various statistics and optimization results

`run_stock_analysis()`

- Main analysis function that orchestrates the workflow
- Shows progress indicators during computation
- Generates all visualizations based on user selections
- Displays results in an organized manner

Data Caching

The application uses Streamlit's `@st.cache_data` decorator to cache the stock data download, reducing API calls and improving responsiveness when parameters change.

Portfolio Optimization Algorithm

The application implements Modern Portfolio Theory through:

1. Monte Carlo simulation of `num_portfolios` random weight combinations
2. Calculation of expected return and volatility for each portfolio
3. Computing the Sharpe ratio (risk-adjusted performance)
4. Finding the portfolio with the maximum Sharpe ratio

Troubleshooting

Common Issues

No Data Downloaded

- **Cause**: Invalid ticker symbols or no data for the selected date range
- **Solution**: Verify ticker symbols use the correct format (e.g., add `.NS` for NSE India stocks) and adjust the date range

Error in Portfolio Calculations

- **Cause**: Insufficient historical data points for reliable statistics
- **Solution**: Extend the date range to include more trading days

Slow Performance

- **Cause**: Too many portfolio simulations or very long date ranges
- **Solution**: Reduce the number of simulations or narrow the date range

Missing Moving Averages

- **Cause**: Date range shorter than the moving average window
- **Solution**: Either extend the date range or reduce the MA window size

References

- **Modern Portfolio Theory**: Markowitz, H.M. (1952). "Portfolio Selection". The Journal of Finance
- **Sharpe Ratio**: Sharpe, W.F. (1966). "Mutual Fund Performance"
- **Technical Analysis**: Murphy, J.J. (1999). "Technical Analysis of the Financial Markets"
- **Data Source**: Yahoo Finance API via yfinance package